FRAs, hedging and beta hedging

Based on Manuel Oyhamburu, UCEMA, Classes 1-7 (2026), with the standard framework taught in John C. Hull, Options, Futures, and Other Derivatives.

A focused guide to understanding the exposure, choosing the correct hedge, calculating its size, and explaining what risk remains. Written in English, with selected Spanish terms to connect it to your slides.

The central question: What market movement hurts my existing position, and what derivative position gains when that happens?

Study route	Guide pages	Your presentation: PDF pages
Hedge direction, basis risk and cross hedging	2-4	21, 26, 31-33, 52-54, 86, 89-90
Beta: neutralize or change market exposure	5-6	87-88, 91
Forward rates and FRA cash flows	7-10	61-64, 69-73, 81-84
Interest-rate hedging and practical limits	11-12	77-80, 85, 89-90
Formula sheet; 18 exercises; solutions	13-19	Original practice based on these topics
Sources and final revision checklist	20	Reading map and source notes

How to study

First pass: read pages 2-12 and reproduce each worked example with a calculator. **Second pass:** use page 13 to solve exercises 1-18 without viewing the answers. **Third pass:** explain every hedge direction aloud before calculating.

Scope and conventions

The focus is forward/futures hedging and FRAs. Duration hedging is included because it connects the bond and Treasury-futures material. Option hedges appear only as a short comparison; advanced option Greeks and swap valuation are outside this guide.

All new market prices and contract inputs are hypothetical. Amounts are in USD unless stated otherwise. Use rates as decimals, and use the stated day count. For numerical futures examples, ignore fees, funding, daily-settlement interest and taxes unless a question says otherwise.

Your Hull edition was not supplied. Chapter references use the publisher's 11th-edition contents; the explanations and exercises are original, not quotations or reproduced textbook solutions.

1 / Understand the hedge

Presentation: PDF pp. 21, 26, 31-33 and 93. Hull reading: Chapters 2-3.

Hedging (cobertura) reduces an exposure you already have, or expect to have. Speculation deliberately takes exposure; arbitrage seeks to exploit inconsistent prices through offsetting transactions. The same futures trade can be a hedge for one person and speculation for another.

Your exposure	What hurts you?	Usual hedge
Own an asset; will sell output	Its price falls	Short futures / sell forward
Need to buy an asset later	Its price rises	Long futures / buy forward
Must pay USD; rate quoted ARS per USD	USD becomes more expensive in ARS	Long USD futures
Will receive USD; measure value in ARS	USD becomes cheaper in ARS	Short USD futures
Will borrow at a floating benchmark	Benchmark interest rate rises	Buy FRA: pay fixed, receive floating
Will invest at a floating benchmark	Benchmark interest rate falls	Sell FRA: receive fixed, pay floating

Read the payoff before memorizing the direction

$$\text{Long futures P\&L} = n \times m \times (F_H - F_0)$$
$$\text{Short futures P\&L} = n \times m \times (F_0 - F_H)$$

Here n is the positive number of contracts, m is units per contract or the monetary multiplier, F_0 is the entry price, and F_H is the price when you close. A long position benefits from a higher quoted futures price. For many interest-rate futures, a higher interest rate means a *lower* quoted futures price.

What success looks like

Evaluate **exposure + derivative**. A futures loss can be evidence that the hedge worked: a wheat producer loses on short futures when wheat rises, but sells its crop at a better price. A hedge usually gives up favorable movements as well as offsetting unfavorable ones.

Forwards can be tailored to the amount and date. Futures are standardized and settled daily. Initial margin is collateral, not the purchase price of the underlying and not the notional amount used to size a hedge. Daily cash requirements can matter even when the combined economic exposure is well hedged.

2 / Basis risk and the effective price

Presentation: PDF pp. 52-54, 89-90. Hull reading: Chapter 3.

Basis at the hedge date: $b_H = S_H - F_H$

S_H is the spot price of the asset being hedged when you end the hedge. The basis **strengthens** when S - F rises, including a move from -5 to -2. It **weakens** when S - F falls. The uncertainty in the ending basis is basis risk.

Short hedge: you will sell the asset

Effective sale price = $S_H + (F_0 - F_H) = F_0 + b_H$

Example: a producer sells futures at 105. At the sale date, spot is 92 and futures are 94. The physical sale earns 92; the short future earns 11. Effective revenue is **103 per unit**, also 105 + (92 - 94).

Long hedge: you will buy the asset

Effective purchase cost = $S_H - (F_H - F_0) = F_0 + b_H$

Example: a buyer goes long at 105. Later, spot is 118 and futures are 117. Futures gain 12, reducing the purchase cost to **106 per unit**. Both formulas give $F_0 + b_H$; one is revenue, the other a cost.

Unexpected basis movement	Short hedger / seller	Long hedger / buyer
Stronger ending basis	Higher effective revenue: helps	Higher effective cost: hurts
Weaker ending basis	Lower effective revenue: hurts	Lower effective cost: helps

These per-unit formulas assume a one-for-one quantity hedge, matching price units and no financing effect from daily settlement. With a different hedge ratio, calculate the physical and derivative cash flows separately.

Why convergence does not solve every hedge

At expiry, futures converge to their **specified underlying or settlement reference**. Your asset may differ in quality, location or composition, so its basis can remain nonzero. If you close before expiry, even an otherwise matched asset can have an uncertain basis.

A standard maturity choice is a liquid contract expiring close to, but after, the hedge date. An earlier expiry requires a roll or creates a coverage gap. Matching an average maturity with two contracts does not automatically minimize risk. Consider delivery rules, liquidity and how each contract co-moves with the exposure.

3 / Cross hedging and the optimal ratio

Presentation: PDF p. 86; interpretation of pp. 89-90. Hull reading: Chapter 3.

A **cross hedge** uses a futures contract on a different asset, such as hedging jet fuel with a related energy future. Correlation helps, but you also need to account for how strongly the two prices move.

Price-change version: use quantities

$$h_{\Delta}^{*} = \mathrm{Cov}(\Delta S, \Delta F) / \mathrm{Var}(\Delta F) = \rho \times \sigma_{\Delta S} / \sigma_{\Delta F}$$
$$n^{*} = h_{\Delta}^{*} \times Q_A / Q_F$$

Q_A is the quantity exposed; Q_F is units per futures contract. The standard deviations are of **price changes** over comparable hedge horizons, not price levels. For a long physical exposure with a short hedge, per-unit change is $\Delta S - h\Delta F$.

$$\mathrm{Var}(\Delta S - h\Delta F) = \sigma_{\Delta S}^{2} + h^{2}\sigma_{\Delta F}^{2} - 2h\,\mathrm{Cov}(\Delta S, \Delta F)$$

Differentiate with respect to h and set the result to zero: $2h\sigma_{\Delta F}^{2} - 2\mathrm{Cov} = 0$. This yields h^{*}. It minimizes variance, not necessarily cost or the probability of every possible loss.

Return version: use monetary values

$$h_{r}^{*} = \mathrm{Cov}(r_A, r_F) / \mathrm{Var}(r_F)$$
$$n^{*} = h_{r}^{*} \times V_A / V_F, \text{ where } V_F = F_0 \times Q_F$$

Here $r_A = \Delta S/S_0$ and $r_F = \Delta F/F_0$. The futures return is the percentage change in the quoted price, **not return on margin**. For fixed starting prices, $h_{r}^{*} = h_{\Delta}^{*} \times F_0/S_0$, so the two methods give the same contract count.

Worked example

A buyer needs 100,000 units. Each futures contract covers 5,000 units. Price-change correlation is 0.80, spot-change volatility is 6 and futures-change volatility is 8. Then $h^{*} = 0.80 \times 6/8 =$ **0.60**; buy $0.60 \times 100{,}000/5{,}000 =$ **12 contracts**.

In the ideal one-factor linear model, the residual variance is $\sigma_{\Delta S}^{2}(1 - \rho^{2})$: here $36 \times 0.36 = 12.96$. The hedge removes 64% of variance. Real performance can differ as correlations and exposures change.

Slide 86 clarification: it displays a price-change ratio next to a monetary-value formula. Pair price changes with quantities, or percentage returns with values. Mixing the two can mis-size the hedge. A negative estimated ratio reverses the usual hedge direction.

4 / What beta hedging removes

Presentation: PDF pp. 87-88. Hull reading: Chapter 3.

Beta (β) measures sensitivity to market returns, not total volatility. A beta of 1.4 means an estimated 1% market move is associated with about a 1.4% portfolio move attributable to the market factor, under a linear model.

$$\beta_P = \mathrm{Cov}(r_P, r_M) / \mathrm{Var}(r_M)$$
$$r_P - r_f = \alpha + \beta_P(r_M - r_f) + \varepsilon$$

r_P is portfolio return; r_M is market return; r_f is the risk-free return over the same period. The market term is systematic risk. ε is the residual from stock selection and other exposures not explained by the market. α is the regression intercept, not a guaranteed excess return.

Portfolio beta uses market-value weights

$$\beta_P = \Sigma\, w_i\beta_i, \text{ with } w_i = \text{market value of holding } i / V_P$$

Example: 40% in a stock with β = 0.8 and 60% in one with β = 1.5 gives $\beta_P = 0.4(0.8) + 0.6(1.5) =$ **1.22**. Do not weight by the number of shares unless share counts happen to produce the same value weights.

Size a hedge or choose a target beta

$$V_F = \text{futures index level} \times \text{currency multiplier per point}$$
$$n_{short} = (\beta_P - \beta_{target}) \times V_P / V_F$$

This standard formula assumes the futures percentage price change tracks the chosen market index with sensitivity approximately one. A positive result means short futures; a negative result means buy the absolute number. To neutralize market exposure, set $\beta_{target} = 0$.

$$\text{After n short contracts: } \beta_{after} \approx \beta_P - nV_F/V_P$$

For a contract with measured futures-price-return sensitivity β_F to that same benchmark, divide the target-beta formula by β_F. This is a factor adjustment, not the beta of the margin account. Alternatively, estimate the hedge directly against futures returns using the return-regression formula on page 4.

Beta-neutral does not mean risk-free

The hedge targets the market component. Stock-specific losses, sector mismatch, changing beta, basis risk and funding costs can remain. A well-diversified perfectly matched hedge may approximate a cash-like return under textbook assumptions; a zero estimated beta alone does not establish that result.

5 / Beta hedging, step by step

Presentation: PDF pp. 87-88 and multiplier example on p. 91. All prices here are hypothetical.

You hold a **$10 million** equity portfolio with β = **1.2**. The index future trades at **5,000** and its multiplier is **$50 per point**. Assume the futures and market percentage changes match for these scenarios.

V_F = 5,000 × $50 = $250,000
Full hedge: n = 1.2 × $10,000,000 / $250,000 = **48 short**

Market / future move	Portfolio market P&L	48-short futures P&L	Combined
-5%; future to 4,750	-$600,000	+$600,000	$0
+5%; future to 5,250	+$600,000	-$600,000	$0

For the down scenario: portfolio market component = $10m × 1.2 × (-0.05). Futures P&L = 48 × $50 × (5,000 - 4,750). These are **market-component** offsets; the table sets residual return, income and financing to zero to isolate the hedge.

Reduce beta without taking it to zero

Target β = 0.5: n = (1.2 - 0.5) × $10m/$250,000 = **28 short**

If the market falls 5%, the original portfolio market loss is $600,000. The short futures gain 28 × $50 × 250 = $350,000. Net market loss is **$250,000**, matching $10m × 0.5 × 5%.

Increase beta

Target β = 1.5: n_{long} = (1.5 - 1.2) × $10m/$250,000 = **12 long**

This adds market exposure. It is a tactical adjustment or exposure increase, rather than protection from a market decline.

An imperfect hedge is not necessarily a failed hedge

Return to the full hedge. Suppose the market falls 5%, but the portfolio falls 8%, including a -2% residual beyond its -6% market component. Stocks lose $800,000; futures gain $600,000. The remaining **$200,000 loss** is the unhedged residual in this example.

Round to tradable whole contracts, then calculate the achieved beta. Reassess after material changes in portfolio value, composition, beta or the futures price. Futures give exposure without paying the full notional, but adverse daily settlements still require cash.

6 / From spot rates to forward rates

Presentation: PDF pp. 61-64 and 69-73. Hull reading: Chapter 4; Chapter 5 for the distinct topic of asset forward prices.

A **spot zero rate** prices borrowing or lending from today to one maturity. A **forward interest rate** is implied today for an interval that starts later. It is a break-even rate from the curve. It is not necessarily the rate that will actually prevail.

Start from discount factors

$$1 + f_0(T_1,T_2)\delta = P(0,T_1) / P(0,T_2)$$
$$f_0(T_1,T_2) = [P(0,T_1)/P(0,T_2) - 1] / \delta$$

P(0,T) is today's price of one unit paid at T. δ is the year fraction of the forward interval. This is the single-curve, simple-rate relation: growth to T_2 must equal growth to T_1 followed by forward growth to T_2.

Match the formula to the rate convention

Input zero-rate convention	Forward relation
Simple annual rates R_1, R_2	$f = [(1 + R_2T_2)/(1 + R_1T_1) - 1] / \delta$
Annual effective zero rates z_1, z_2	$f_{eff} = [(1+z_2)^{T_2}/(1+z_1)^{T_1}]^{1/(T_2-T_1)} - 1$
Continuously compounded zero rates R_1, R_2	$f_c = (R_2T_2 - R_1T_1)/(T_2-T_1)$

For the simple formula, T_1 and T_2 must use compatible day-count fractions. For continuous-to-simple conversion over a consistent year interval δ: $\mathbf{f_{simple} = [\exp(f_c\delta) - 1]/\delta}$. Never insert a continuous rate directly into a simple-interest FRA payoff.

Two quick examples

Slide 63: one-year annual effective zero rate 5%; two-year rate 6%. The one-year forward beginning in year one is $1.06^2/1.05 - 1$ = **7.0095%**, shown rounded to 7% in the slide.

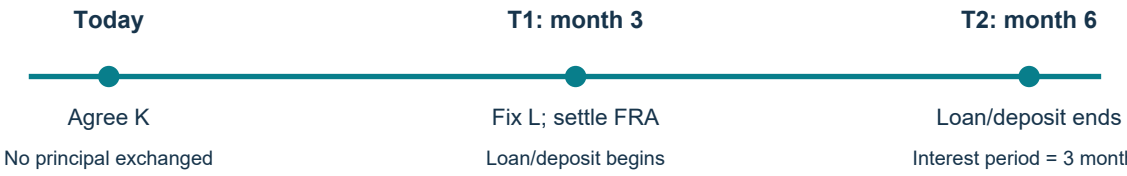
Simple-rate example: 3-month zero rate 4%, 6-month zero rate 5%, with $T_1 = 0.25$ and $T_2 = 0.50$. The 3×6 forward is $[(1.025/1.01) - 1]/0.25$ = **5.9406% p.a.** This is the fair FRA fixed rate in the matching single-curve model.

Do not confuse this f with an asset forward price F_0, such as the number of ARS per USD. No-arbitrage identifies an implied rate. Equating it with the expected future spot rate requires additional expectations assumptions; risk premia can drive a wedge between them.

7 / FRA mechanics and the timeline

Presentation: PDF pp. 81-83. Standard settlement convention also checked against ISDA disclosure [3] and HMRC [4].

A **forward rate agreement** fixes an interest rate K today for a future accrual period on a notional amount. This guide uses **N for notional** and **L for the observed floating reference rate**. No principal is lent through the FRA itself.



A **3×6 FRA** starts in three months and covers the interval from month 3 to month 6: three months of interest. A 5×8 FRA starts in five months and also covers three months. The second number is the end date measured from today, not the length of the loan.

Side, following your slides	Economic exchange	Who usually uses it?
Long / FRA buyer	Pay fixed K; receive floating L	Future borrower protecting against higher rates
Short / FRA seller	Receive fixed K; pay floating L	Future depositor protecting against lower rates

Three dates with three different jobs

Trade date: agree N, K, the currency, reference rate, start/end dates and day-count convention. An at-market FRA has approximately zero initial value, before fees and credit/funding effects.

T_1, start of coverage: the reference rate fixes at the contractual fixing time, often shortly before T_1. Under the conventional FRA in your slides, the net payment occurs at T_1.

T_2, end of coverage: the separate underlying loan or deposit finishes. The conventional FRA has already settled; do not add another derivative payment at T_2.

What does the FRA actually lock?

It hedges the named reference rate over the named interval. If your borrowing rate is L + s with a fixed spread s, the combined rate is K + s under matching cash-flow assumptions. If s changes, or your actual borrowing uses another benchmark or tenor, that extra risk remains. The FRA also does not guarantee access to the loan principal.

A swap can be understood as a sequence of fixed-versus-floating interest periods. A single FRA covers one period; a series of FRAs can hedge a sequence, with each date and notional matched to the exposure.

8 / FRA settlement: follow the cash

Presentation: PDF pp. 82-84. The example below reconstructs p. 84 with explicit signs and cash-flow dates.

Buyer's equivalent interest difference at T_2 = $N(L - K)\delta$
Buyer's cash settlement at T_1: $C_1 = N(L - K)\delta / (1 + L\delta)$
Seller's cash settlement = $-C_1$

Positive C_1 means the buyer receives. Negative C_1 means the buyer pays. δ = actual days/360 or actual days/365 as specified. The denominator brings the **interest difference** back from T_2 to T_1; principal is not exchanged.

Why divide by 1 + Lδ?

If the buyer receives C_1 at T_1 and invests it at L for the period, it becomes $C_1(1 + L\delta) = N(L - K)\delta$ at T_2. Thus the early payment is economically equivalent to the end-period interest difference, assuming the relevant reinvestment/financing rate is L.

Slide 84: N = $10m, K = 4%, L = 5%, 90/360

δ = 0.25
Difference at T_2 = $10m × (0.05 - 0.04) × 0.25 = $25,000
C_1 = $25,000 / 1.0125 = **$24,691.36 received by buyer**

The $125,000 displayed in that slide is total floating interest, $10m × 5% × 0.25. Fixed interest is $100,000. Their difference is **$25,000**, which is the amount to discount. The settlement is neither $125,000 nor the present value of $10 million.

Prove that it hedges a borrower

Reference rate at T_1	Loan interest at T_2	FRA amount carried to T_2	Net interest cost
5%	Pay $125,000	Receive $25,000	$100,000
3%	Pay $75,000	Pay $25,000	$100,000

At L = 3%, the buyer pays $25,000/1.0075 = **$24,813.90** at T_1. Financing that payment at L produces a $25,000 cost at T_2. The favorable lower borrowing rate is offset by the FRA loss; the combined benchmark cost remains 4%.

An actual borrowing spread, unmatched funding rates, dates, day counts or notionals can prevent this exact offset. For the exam, state the matching assumptions before claiming a locked all-in rate.

9 / Pricing versus valuing an FRA

Hull framework: Chapter 4. Single-curve classroom formulas are distinguished below from modern projection/discounting practice.

Pricing at initiation asks which fixed rate K gives zero initial value. **Valuing an existing FRA** asks what an old contract is worth after market forward rates change. K stays fixed; the current forward rate changes.

Single-curve valuation before the fixing date

Fair initial rate: $K = f_0(T_1,T_2)$
Buyer value at t: $V_t = N\delta[f_t(T_1,T_2) - K]P(t,T_2)$

Here $t < T_1$, and f_t is today's implied simple rate for the **original** accrual interval. In a consistent single-curve model $P(t,T_1)/(1 + f_t\delta) = P(t,T_2)$, so an equivalent expression is:

$$V_t = \{N\delta(f_t - K)/(1 + f_t\delta)\} \times P(t,T_1)$$

This separates the start-date settlement equivalent from discounting back to the valuation date. Do not use a spot zero rate in place of the forward rate for the covered interval. Do not discount the same time interval twice.

Worked valuation

An existing long FRA has N = \$10m, K = 4%, δ = 0.25. Before fixing, the new forward for the original interval is 5.2%, and $P(t,T_2) = 0.98$.

V_t = \$10m × 0.25 × (0.052 - 0.04) × 0.98
= **+\$29,400 to the buyer**; seller value = -\$29,400

The contract is valuable to the buyer because its fixed borrowing benchmark is below the current forward. This is a **current value**, not a known future settlement: L has not fixed yet.

Recognize the model boundary

Modern valuation may use one curve to project the reference rate and another, typically collateral/OIS-based, curve to discount cash flows. For a payoff $N\delta(L - K)$ actually paid at T_2, the familiar discounted-forward form applies with a consistent projection and discounting framework.

For a conventional **start-settled** FRA, the actual payoff contains L in the denominator. Outside the single-curve assumptions, do not automatically substitute a projected forward into that nonlinear payoff and call it exact; settlement conventions and possible convexity effects matter. The numerical exercises here deliberately use the single-curve model.

After settlement at T_1, the settled FRA has no remaining payment and hence no remaining value, even though the separate loan/deposit continues to T_2.

10 / Hedging interest-rate exposure

Presentation: PDF pp. 77-80 and 85. Hull reading: Chapter 6. Extension: duration/DV01 sizing connects the slide topics.

Risk to hedge	FRA position	Price-quoted rate/bond futures
Future borrowing: rates rise	Long: pay fixed	Short: prices fall as rates rise
Future deposit: rates fall	Short: receive fixed	Long: prices rise as rates fall
Existing fixed-rate bonds: yields rise	Match the specific rate exposure; one FRA may be inadequate	Short suitable bond futures

A long FRA gains when the reference rate rises. A long bond-price future generally loses when yields rise. For a short-term contract quoted as **100 minus the rate in percent**, a rise from 4.0% to 4.5% lowers the quote from 96.0 to 95.5. Always check the quote convention; a directly yield-quoted contract reverses this price-quoted direction.

Duration and DV01

$$\Delta V \approx -D_{mod} \times V \times \Delta y$$
$$DV01 \approx D_{mod} \times V \times 0.0001$$
$$n_{short} = DV01_{portfolio} / DV01_{one\ long\ futures\ contract}$$

D_{mod} is modified duration; Δy is a decimal yield change. Here DV01 is the positive approximate dollar loss for a 1 bp rise on an ordinary long bond position. One basis point is 0.01 percentage point = 0.0001 in decimals. For periodic yield compounding, $D_{mod} = D_{Macaulay}/(1 + y/m)$, with m payments per year.

Example: a $5m bond portfolio has modified duration 4. Its DV01 is 4 × $5m × 0.0001 = **$2,000/bp**. If one long futures contract has DV01 $80/bp, short **25 contracts**. A parallel +10 bp move gives about -$20,000 on bonds and +$20,000 on the futures hedge.

What this approximation leaves behind

Duration is a first-order approximation. Convexity matters for larger yield changes; curve twists need maturity-specific sensitivities. Credit spreads may move differently from the government benchmark. Deliverable bond futures also involve cheapest-to-deliver and conversion-factor effects: use the futures contract's appropriate DV01, not its face amount alone.

A rate hedge can be sized similarly by matching dollars per basis point. For one future interest period, the undiscounted exposure is Nδ × 0.0001 per bp. FRAs and daily-settled interest-rate futures have different timing and may need a convexity adjustment; they are not automatically interchangeable at the same quoted rate.

11 / Build and evaluate a complete hedge

Presentation: PDF pp. 21, 31-33, 86 and 89-90. Related option contrast: pp. 94-100.

A six-step exam method

1. Describe the exposure. State the asset, amount, cash-flow date, currency and adverse market move.
2. Select the instrument. Match the underlying/reference rate, tenor and settlement conventions.
3. Choose direction. Explain which side gains in the adverse scenario.
4. Size it. Use matched quantity, a regression hedge ratio, beta or DV01.
5. Combine cash flows. Put them in the same currency and at the same date.
6. State what remains. Identify basis, residual, amount, date or financing mismatches.

Practical risks worth naming

Basis and correlation: the hedging instrument may move differently. **Quantity:** uncertain production or a cancelled purchase can turn a hedge into an open position. **Rollover:** closing one maturity and opening another exposes you to the calendar spread. **Liquidity:** margin payments can arrive before the offsetting business cash flow. **Counterparty:** an OTC counterparty may fail to pay. **Model:** historical beta and covariance estimates can change.

Assess the business's **net** exposure. If higher input costs are automatically passed into selling prices, hedging just the input may increase business-margin variability. Hedging can stabilize cash flows and reduce funding pressure, but also has costs.

A concise comparison with option protection

Instrument	Economic effect	Tradeoff
Short future against an owned asset	Approximately fixes sale economics	Offsets favorable price increases too
Protective put against an owned asset	Creates an expiry floor on the matched asset	Pay premium; retains upside
Call against a future purchase	Creates an expiry ceiling on matched purchase cost	Pay premium; retains benefit of lower prices
FRA against one floating interest period	Fixes the reference-rate component	Gains and losses in both directions

One asset plus a matched put has expiry value $\max(S_T, K)$, before premium. Deduct the premium's future value for an expiry-date comparison. Beta measures broad market sensitivity; option delta measures local sensitivity to the underlying.

Slide clarifications to remember

A known nonzero basis can permit a fixed outcome. Modest correlation can still reduce variance. A hedge ratio or beta is not a correlation coefficient. Calendar distance alone does not determine the best hedge maturity.

12 / Formula sheet

Use this page only after you can explain the exposure and the sign of the hedge.

Task	Formula and convention
Futures P&L	Long: $n\,m(F_H-F_0)$. Short: $n\,m(F_0-F_H)$. n is positive.
Basis	$b = S - F$. Strengthening = an increase in b.
One-for-one effective price	Seller revenue or buyer cost = $F_0 + b_H$.
Minimum-variance ratio: price changes	$h_\Delta^* = \rho\,\sigma_{\Delta S}/\sigma_{\Delta F}$; $n^* = h_\Delta^* Q_A/Q_F$.
Minimum-variance ratio: returns	$h_r^* = \mathrm{Cov}(r_A,r_F)/\mathrm{Var}(r_F)$; $n^* = h_r^* V_A/V_F$.
Portfolio beta	$\beta_P = \Sigma w_i\beta_i$; w_i are market-value weights.
Index futures notional	$V_F = F_0 \times$ monetary multiplier per point.
Target beta; futures beta ≈ 1	$n_{short} = (\beta_P-\beta_{target})V_P/V_F$. Negative means long.
Achieved beta after short hedge	$\beta_{after} \approx \beta_P - nV_F/V_P$.
Simple forward interest rate	$f = [P(0,T_1)/P(0,T_2)-1]/\delta$, in a single-curve model.
Continuous forward interest rate	$f_c = (R_2T_2-R_1T_1)/(T_2-T_1)$. Convert before simple FRA use.
FRA settlement to buyer at T_1	$N(L-K)\delta/(1+L\delta)$. Positive: receive; negative: pay.
Pre-fixing FRA value; single curve	$N\delta(f_t-K)P(t,T_2)$ to buyer; opposite to seller.
Duration / DV01 hedge	$\Delta V \approx -D_{mod}V\Delta y$; $DV01 \approx D_{mod}V\times0.0001$; short $n = DV01_P/DV01_F$.

Symbols: n = futures count; N = FRA notional; m = futures multiplier; δ = FRA accrual year fraction; L = realized reference rate; K = FRA fixed rate; f_t = current forward interest rate; $P(t,T)$ = discount factor; V = monetary value; Q = physical quantity.

Before submitting: rates in decimals; correct year fraction; no principal exchange in FRA; price changes versus returns consistent; multiplier included; same cash-flow date; whole-contract rounding checked; remaining risk named.

Practice A / Forward rates and FRAs

Exercises 1-6. Suggested time: 35 minutes. Solutions begin on guide p. 17. Use the stated single-curve assumptions.

1. Read the contract - foundation

You must borrow $8m in six months for three months. (a) Name the FRA using a×b notation. (b) Buy or sell it? (c) When does the conventional FRA settle? (d) Does the FRA supply the $8m principal?

2. Price a forward rate - calculation

The 3-month and 6-month simple annual zero rates are 4.8% and 5.4%. Use $T_1 = 0.25$, $T_2 = 0.50$ and $\delta = 0.25$. (a) Calculate the fair 3×6 FRA rate. (b) Explain the no-arbitrage growth relation. (c) Is the result a guaranteed prediction?

3. Settle the buyer's FRA - calculation

A long FRA has N = $5m, K = 6%, and a 92-day accrual period on Actual/360. At fixing L = 7.2%. Calculate (a) the equivalent T_2 interest difference and (b) the T_1 settlement, including who pays whom. (c) Why are the two amounts different?

4. Hedge a future deposit - application

A company will deposit $12m for 90 days on Actual/360 and sells an FRA at K = 5%. The reference rate fixes at 3.8%. (a) Compute its FRA receipt at T_1. (b) Compute deposit interest at T_2. (c) Carry the FRA receipt forward at 3.8% and show the combined effective annual rate.

5. Value an existing FRA - application

Before fixing, an existing long FRA has N = $20m, K = 4.5% and $\delta = 0.25$. The current forward for the same original period is 5.1%; $P(t,T_2) = 0.975$. (a) Value both sides using the single-curve model. (b) Does this amount have to equal the eventual settlement? (c) Does K change?

6. Compounding and the borrowing spread - challenge

(a) Continuous zero rates are 4% to 0.5 years and 4.6% to 1 year. Find the continuous forward for the half-year interval and its simple annual equivalent using $\delta = 0.5$. (b) Separately, a borrower pays L + 1.5% on a $10m, 90/360 loan and buys a matching FRA at K = 4%. If L fixes at 6%, find the FRA payment at T_1 and the net T_2 interest cost, carrying the FRA cash flow at L. What changes if the spread rises to 2%?

Show your work: timeline → exposure → direction → convention → formula → amount and payer → economic interpretation.

Practice B / Futures and cross hedging

Exercises 7-12. Suggested time: 35 minutes. Assume positive price correlation unless stated otherwise. Solutions on p. 18.

7. Choose the direction - foundation

Choose long or short futures and explain why: (a) a farmer will sell grain; (b) a manufacturer will buy copper; (c) an Argentine importer must pay USD, with futures quoted ARS/USD; (d) an exporter will receive USD under the same quote. (e) For a future borrower, compare a long FRA with a future quoted 100 minus the reference rate.

8. Short hedge and basis - calculation

A producer will sell 50,000 units and sells 10 futures, each covering 5,000 units, at F_0 = \$82 per unit. At the sale date S_H = \$74 and F_H = \$76. Find the physical revenue, futures P&L, ending basis, total revenue and effective sale price. If the ending basis had been \$1 instead, what would the effective price have been?

9. Long hedge - calculation

A buyer needs 30,000 units and buys 6 contracts of 5,000 units each at \$100. At purchase, spot is \$113 and futures are \$110. Find futures P&L, gross physical purchase cost, net cost and effective cost per unit. Explain why it did not lock \$100.

10. Price-change cross hedge - application

A business expects to buy 200,000 units. Each futures contract covers 10,000 units. Over comparable periods, $\rho = 0.75$, $\sigma_{\Delta S}$ = \$4 and $\sigma_{\Delta F}$ = \$5 per unit. Find the optimal ratio, direction and contract count. What fraction of unhedged variance remains in the ideal linear model?

11. Return-based sizing and rounding - challenge

An owned asset position is worth \$3m. The futures quote is \$60 per unit and each contract covers 1,000 units. Return correlation is 0.80; asset-return volatility is 12%; futures-price-return volatility is 15%. (a) Find h_r^* and the contract count. (b) Now use a price-change ratio $h_\Delta^* = 0.70$, Q_A = 123,000 and Q_F = 5,000. Find the theoretical count and the whole number closest to the variance-minimizing solution. (c) Why is blindly mixing the two conventions incorrect?

12. DV01 and practical risk - application

A \$7.5m bond portfolio has modified duration 3.2. One long bond futures contract has DV01 \$60/bp. (a) Size a hedge against rising yields. (b) Estimate the bond loss and futures gain for a parallel +15 bp shift. (c) Name two risks that this hedge leaves. (d) Why can a short-futures hedger face a cash shortage in a rising market?

Practice C / Beta hedging

Exercises 13-18. Suggested time: 40 minutes. Futures-price-return beta is one; ignore income, financing and residual return except where specified. Solutions on p. 19.

13. Calculate portfolio beta - foundation

A $10m portfolio contains $2m of A (β = 0.7), $3m of B (β = 1.1), and $5m of C (β = 1.5). Calculate the weights and portfolio beta. Explain why share-count weights would usually be wrong.

14. Remove market exposure - calculation

Use the portfolio in Exercise 13. The index future trades at 4,000 with a $50 multiplier. (a) Find each contract's notional and the full beta hedge. (b) If both the market index and future fall 4%, calculate the portfolio's market-component loss and the hedge gain.

15. Change the target beta - application

A portfolio is worth $15m and has β = 1.3. Futures trade at 5,000 with a $50 multiplier. (a) Target β = 0.4: number and direction? (b) Starting again with no futures, target β = 1.6: number and direction? (c) Which trade increases market exposure?

16. Residual risk - application

A $10m portfolio has β = 1.2 and is fully beta hedged using 48 short futures entered at 5,000, multiplier $50. The market and futures fall 5%, but the actual portfolio return is -9%. Find the stock P&L, hedge P&L and net P&L. Identify the residual return. Does a remaining loss prove that market hedging was incorrectly sized?

17. Whole contracts and achieved beta - challenge

An $8.3m portfolio has β = 1.17. Futures trade at 4,850 with a $50 multiplier. Target β = 0.30. (a) Find the fractional short count. (b) Compare achieved betas using 29 and 30 contracts. Which is closer to the target? (c) Why might a manager deliberately choose the other integer?

18. Combine two distinct hedges - challenge

A firm holds a $20m equity portfolio with β = 1.1 and wants β = 0.2. Futures trade at 5,000 with multiplier $50. Separately, in three months it must borrow $6m for 90/360 at L + a fixed 2% spread; it enters a matching FRA at K = 4.8%. (a) Specify both hedges. (b) If the market/future fall 3% and L fixes at 6%, calculate the equity market loss after hedging, the FRA settlement at T_1, and the loan's net T_2 interest cost. Carry FRA cash at L. (c) Explain why these two results should not be added without aligning cash-flow dates and definitions.

Solutions A / FRAs

Answers to Exercises 1-6. Positive FRA settlement/value is stated from the buyer's perspective unless indicated otherwise.

1. 6×9; buy the FRA

The interval begins in month 6 and ends in month 9. Buy/pay fixed to offset rising borrowing rates. Conventional cash settlement is at month 6, based on the contractual fixing. The principal comes from a separate loan, not from the FRA.

2. Fair rate = 5.9289% p.a.

f = [(1 + 0.054×0.50)/(1 + 0.048×0.25) - 1]/0.25 = (1.027/1.012 - 1)/0.25 = 0.05928854. The replication condition is 1.027 = 1.012(1 + f×0.25). This is a no-arbitrage implied rate, not a guaranteed forecast.

3. Buyer receives $15,056.30 at T_1

δ = 92/360. T_2 difference = $5m×(0.072-0.06)×92/360 = $15,333.33. Divide by 1 + 0.072×92/360 = 1.0184 to obtain $15,056.30. The seller pays the buyer because L > K. The earlier cash payment is smaller because it can earn interest before T_2.

4. Seller receives $35,661.22 at T_1

Seller amount = $12m×(0.05-0.038)×0.25/(1+0.038×0.25) = $36,000/1.0095 = $35,661.22. Deposit interest = $114,000. The FRA receipt grows to $36,000, so total T_2 interest = $150,000. Annual rate = $150,000/($12m×0.25) = 5%.

5. Buyer value = +$29,250

V = $20m×0.25×(0.051-0.045)×0.975 = $29,250; seller value = -$29,250. This is the current model value, not the eventual cash settlement. The realized fixing can change it. K remains 4.5%; the contract's fixed rate is not reset to the new forward.

6. Continuous versus simple; hedge the benchmark

(a) f_c = (0.046×1 - 0.04×0.5)/0.5 = 5.2%. Simple equivalent = [exp(0.052×0.5)-1]/0.5 = **5.2682%** p.a.
(b) Buyer receipt = $10m×(0.06-0.04)×0.25/1.015 = **$49,261.08**. It grows to $50,000 at T_2. Loan interest = $10m×7.5%×0.25 = $187,500. Net = $137,500, equivalent to **5.5% = K + 1.5%**. If the spread rises to 2%, net cost becomes $150,000, or 6%. The FRA does not hedge the extra 0.5% spread.

Solutions B / Futures hedging

Answers to Exercises 7-12. Futures cash flows are combined without daily-settlement financing effects, as specified.

7. Use the adverse price move

(a) Short grain: falling prices hurt the sale. (b) Long copper: rising prices hurt the purchase. (c) Long USD: a higher ARS/USD rate increases the importer's cost. (d) Short USD: a lower rate reduces the exporter's ARS revenue. (e) Buy the FRA, but short a 100-minus-rate future: both positions gain as the reference rate rises.

8. Effective sale price = $80

Physical revenue = 50,000×$74 = $3,700,000. Futures gain = 10×5,000×($82-$76) = $300,000. Basis = $74-$76 = -$2. Total = $4m; effective revenue = $80/unit = $82-$2. If ending basis were +$1, effective revenue would be $83/unit. A stronger basis helps this seller.

9. Effective purchase cost = $103

Long futures gain = 6×5,000×($110-$100) = $300,000. Physical cost = 30,000×$113 = $3,390,000. Net = $3,090,000, or $103/unit. Ending basis is +$3, so cost = $F_0 + b_H$ = $103. Uncertain ending basis prevents a guaranteed $100 cost.

10. Buy 12 futures

h* = 0.75×4/5 = 0.60. Count = 0.60×200,000/10,000 = 12. Buy because the business is exposed to a future purchase-price increase. Residual variance fraction = 1-0.75² = **43.75%**; the modeled variance reduction is 56.25%.

11. 32 short; then 17 nearest whole contracts

(a) h_r^* = 0.80×0.12/0.15 = 0.64. V_F = $60×1,000 = $60,000. Count = 0.64×$3m/$60,000 = **32 short** against the owned asset.
(b) n* = 0.70×123,000/5,000 = 17.22; 17 is nearest under the stated variance objective.
(c) A price-change slope scales quantities; a return slope scales monetary exposure. Price levels convert one slope into the other. Without the conversion, equal-looking numbers need not represent the same hedge.

12. Short 40 contracts

$DV01_P$ = 3.2×$7.5m×0.0001 = $2,400/bp. Count = $2,400/$60 = **40 short**. A +15 bp shift gives about -$36,000 on bonds and +$36,000 on futures. Curve twists, convexity, credit spreads or a changing cheapest-to-deliver bond can leave risk. A rising futures price requires cash payments by a short position; gains on the underlying may remain unrealized or arrive later.

Solutions C / Beta hedging

Answers to Exercises 13-18. The net results isolate the components requested; zero beta is not a promise of zero total return.

13. Portfolio beta = 1.22

Weights are 20%, 30%, 50%. β = 0.20×0.7 + 0.30×1.1 + 0.50×1.5 = 1.22. Money invested determines each holding's contribution to portfolio return; share numbers ignore different share prices.

14. Short 61 contracts

V_F = 4,000×\$50 = \$200,000. n = 1.22×\$10m/\$200,000 = 61. Market-component stock loss = \$10m×1.22×4% = \$488,000. Futures fall 160 points; gain = 61×\$50×160 = \$488,000. Modeled market component nets to zero.

15. 54 short; or 18 long

V_F = \$250,000. (a) (1.3-0.4)×\$15m/\$250,000 = **54 short**. (b) (1.6-1.3)×\$15m/\$250,000 = **18 long**, starting from no futures as asked. (c) The long position increases beta and market exposure.

16. Residual loss = \$300,000

Stocks lose \$900,000. Futures gain 48×\$50×250 = \$600,000. Net P&L = **-\$300,000**. The beta-predicted return is -6%; actual return is -9%, leaving a -3% residual. Correct market-factor sizing can coexist with a remaining stock-specific or otherwise unexplained loss.

17. 30 contracts is closer

V_F = 4,850×\$50 = \$242,500. n* = (1.17-0.30)×\$8.3m/\$242,500 = **29.7773 short**.
29 contracts: β ≈ 1.17 - 29×242,500/8,300,000 = **0.3227**.
30 contracts: β ≈ **0.2935**, closer to 0.30. A policy forbidding beta below 0.30 could justify choosing 29, even though it is farther from the point target.

18. 72 short index futures; buy a 3×6 FRA

(a) Equity hedge = (1.1-0.2)×\$20m/\$250,000 = **72 short**. Buy/pay fixed on the \$6m 3×6 FRA.
(b) Stocks' market loss = \$20m×1.1×3% = \$660,000; futures gain = 72×\$50×150 = \$540,000. Net market loss = **\$120,000**, matching beta 0.2. FRA receipt at T_1 = \$6m×(0.06-0.048)×0.25/1.015 = **\$17,733.99**. It grows to \$18,000 at T_2. Loan interest = \$6m×8%×0.25 = \$120,000; net = **\$102,000**, or 6.8% = 4.8% + 2%.
(c) The equity number is a value change, the FRA number is an early cash receipt, and the loan number is an end-period interest cost. State valuation dates and bring cash flows to a common date before aggregating.

Sources and final revision

PDF page references count the cover as page 1. External sources checked September 13, 2026.

Primary course material

[1] Manuel Oyhamburu. *Forwards y Futuros*, UCEMA, Classes 1-7, 2026. User-provided file: PPT 1_Futuros_Manuel Oyhamburu 2026_UCEMA_Clases 1 a 7.pdf. Core reading: pp. 52-54 (basis), 61-64 (forward rates), 77-80 (duration), 81-84 (FRAs), 85-90 (rate futures and hedging), 91 (index multiplier).

Hull reading map

[2] John C. Hull. *Options, Futures, and Other Derivatives*, 11th edition, Pearson. Read Chapter 3 for futures hedge strategy, cross hedges and beta; Chapter 4 for interest rates and FRAs; Chapter 6 for interest-rate futures and duration hedging. Chapters 2 and 5 provide mechanics and no-arbitrage background. The edition's contents were verified on the publisher's page; your full textbook was not provided or directly inspected.

Pearson: Hull, 11th edition, overview and table of contents

Supplementary checks

[3] ISDA. *Interest Rate Derivatives Disclosure Annex*, hosted by Commonwealth Bank. The FRA section supports discounting interest differences when settlement precedes accrual-period end. Used to verify the settlement convention, not as a source of the guide's exercises.

ISDA disclosure annex: interest-rate derivatives

[4] HMRC. *Corporate Finance Manual, CFM13150.* Supports the notional-loan interpretation, a single-period fixed/floating exchange and conventional early settlement. Its legacy reference-rate example is not used as a current market specification.

HMRC: forward rate agreements

[5] CME Group. *Hedging and Risk Management for Equity Index Futures.* Used to verify contract notional as index level times multiplier and the mechanics of the offsetting futures cash flow. Numerical exercises in this guide are original.

CME: equity index futures hedging

You are ready when you can explain these without notes

Why a borrower buys an FRA but shorts a price-quoted interest-rate future. Why an FRA payment is discounted to T_1. Why a short hedge benefits from a stronger basis. Why a price-change hedge ratio uses quantities. Why beta-neutral can still lose money. Why a margin requirement is not contract notional.

All explanations are study material. Follow the rate, day-count and settlement assumptions stated in each exam problem; do not silently combine conventions from different examples.